Business Summary Report

Task 3 — Predictive Insights for Collections Strategy | Geldium

Prepared as a concise business report

1. Summary of Predictive Insights

The purpose of the proposed predictive model is to identify customers who may be at higher risk of delinquency so the Collections team can offer timely, proportionate support. Logistic Regression is the proposed model because its probability estimates and coefficients can be explained to business stakeholders.

The following are candidate risk factors that must be validated against the actual dataset and Task 2 model output before being described as confirmed findings:

• Previous missed or late payments: repeated missed payments may indicate increased risk.

• Credit utilization: a high balance relative to available credit may be associated with repayment pressure.

• Affordability indicators, such as debt-to-income ratio or outstanding balance: include these only if the fields exist and the analysis supports the relationship.

These are hypotheses, not confirmed model findings. The final report should replace them with the actual top predictors and measured patterns from the analysis.

2. Recommendation Framework

Restated insight: If the validated analysis identifies a higher-risk group based on relevant repayment or credit indicators, the Collections team can test an early, supportive reminder intervention.

Proposed SMART recommendation:

• Specific: Run a six-week pilot of respectful SMS payment reminders for customers flagged as high risk by the validated model. Include a clear route to contact Collections for assistance.

• Measurable: Compare the 30-day delinquency rate between the reminder group and a suitable comparison group. Also track repayment outcomes, contact costs, complaints, and opt-outs.

• Actionable: Use an approved messaging process, define eligibility rules, and have Collections staff review the proposed contact list before messages are sent.

• Relevant: The pilot aims to encourage timely payments and offer support before delinquency worsens.

• Time-bound: Review results at the end of six weeks before deciding whether to expand, adjust, or stop the pilot.

Justification and business rationale: A limited pilot allows Geldium to measure effectiveness and cost, monitor customer response, and identify unintended effects before scaling. No specific improvement is assumed; targets should be set after establishing a baseline.

3. Ethical and Responsible AI Considerations

• Bias and fairness: Historical data may reflect unequal past treatment or underrepresent some customer groups. Where lawful and appropriate, compare error rates and outreach rates across relevant groups, investigate disparities, and avoid unjustified sensitive or proxy variables.

• Explainability: Logistic Regression can provide relatively interpretable coefficients and risk estimates. Communicate the main contributing factors in plain language and explain that a risk score is an estimate, not a certainty.

• Responsible decisions: Use predictions to prioritize supportive outreach, not to automatically punish customers or deny options. Keep human review for disputes, hardship requests, exceptions, and consequential decisions.

• Privacy and accountability: Use only authorized, necessary data; restrict access; document model versions and decisions; and monitor model performance and fairness over time.

Validation note: This draft is based on the Task 3 brief and existing project draft. Candidate risk factors are not verified against actual model outputs. Confirm them before submitting.